



RIVER VALLEY HIGH SCHOOL
General Certificate of Education Advanced Level
Higher 2
JC2 Common Test 3

COMPUTING

Paper 1

9569/01**30 JUN 2022****3 hours**

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your center number, index number and name on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use an HB pencil for any diagrams, graphs, tables or rough working.

Do not use staples, paper clips, glue or correction fluid.

DO **NOT** WRITE IN ANY BARCODES.

If you need additional answer paper ask the invigilator for a continuation booklet.

Answer **all** questions.

Approved calculators are allowed.

The number of marks is given in brackets [] at the end of each question or part question.

The total number of marks for this paper is 100.

This document consists of 7 printed pages.

Answer **all** questions.

1. The following conditions are applied to determine the prize of Garapon Lottery Wheel. In each attempt, participant draws 3 balls one at a time. The colours of the balls in the lottery wheel are gold, red, blue and white. Draw a reduced decision table based on the conditions given below.
 - A. If a golden ball is drawn, the grand prize will be awarded.
 - B. If at least one consecutive draw is of the same colour, a \$20 voucher will be awarded.
 - C. If ~~there is fewer than three colours~~ runs out of ball in one attempt, a \$50 voucher will be awarded.
 - D. The grand prize is also awarded if B and C are satisfied. [6]
2. A student wrote a `delete` function in Python.

```
def delete(lst, item):
    if lst[0] == item:
        return lst[1:]
    else:
        return [lst[0],] + delete(lst[1:], item)
```

- a) State the output of the function when the following test case is run.
`print(delete([9, 4, 6, 2, 7, 3, 1, 8, 5], 3))` [1]
- b) Identify the error when the following test case is run.
`print(delete([9, 4, 6, 2, 7, 1, 8, 5], 3))` [1]
- c) Modify the code to correct the error in **b**). [1]
- d) Modify the code further so that the function can delete all occurrences of `item` in the `lst`. [1]

A student wrote the following program in pseudo-code.

```
INPUT n
i = 2
answer = prime
WHILE i <= n / 2
    rem = n % i
    IF rem is not equal to 0
        i = i + 1
    ELSE
        answer = not prime
        BREAK
END WHILE LOOP
OUTPUT answer
```

- e) Draw a flow chart for the pseudo-code above. [5]

3. Computer network and security

- a) As the network administrator of a boarding school, you are tasked to restrict boarders from using the school LAN resources to access online gambling website (e.g. <https://www.singaporepools.com.sg/>) and the use of insecure file transfer protocol (e.g. FTP). Explain how you can achieve this. [2]
- b) What is the purpose of a network switch? [2]
- c) State the most important protocol in the Internet layer, explain how it achieves its goal. [3]
- d) What is circuit switching? [1]
- e) Explain how encryption helps in network security. [2]

4. Hummingway is a music school providing music lessons for young children. The music school engages you to implement an object-oriented solution to keep track of staff information. All staff should have their name, contact and bank account stored.

Some staff belong to the Contract Staff category. Their salary is calculated based on an hourly rate. The **records** of their working hours are stored in a list of tuples, each tuple contains the date of the lesson, and the duration of the lesson.

e.g. [("20220506", 1.5), ("20220507", 1.0), ...]

Some staff belong to the Permanent Staff category. Their salary is calculated based on a monthly salary scheme. In addition, each year they might be given a percentage bonus based on their performance. The bonus should be set as 0.0 by default.

- a) Draw a class diagram, with base class **Staff**, showing:
 - appropriate sub-classes,
 - inheritance,
 - the properties required,
 - appropriate methods, including but not limited to the **constructor** methods, and at least **one** pair of '**get**' and '**set**' methods for each class,
 - circle the polymorphed methods.

[6]
- b) Using the above example, explain the meaning of the term **Encapsulation**, and state one advantage of encapsulation. [3]
- c) Using the above example, explain the meaning of the term **Inheritance**, and state one advantage of inheritance. [3]

- d)** The function takes in the records as a list of tuples, the hourly rate, start date and end date as inputs. It supposed to calculate the salary earned by the contract staff within this period.

```

01 FUNCTION CALC_SALARY (RECORDS: List, HOURLY_RATE: Float,
02                        START_DATE: STR, END_DATE: STR):
03     DECLARE TOTAL_DURATION: INT = 0
04     FOR RECORD IN RECORDS:
05         DATE: STR = RECORD(0)
06         DURATION: INT = RECORD(1)
07         IF START_DATE <= DATE <= END_DATE:
08             TOTAL_DURATION += DURATION
09         END IF
10     END FOR
11     RETURN TOTAL_DURATION * HOURLY_RATE
12 END FUNCTION

```

Identify three types of errors from the above code, state the type and definition of each error and explain how they can be fixed. [6]

- e)** Describe and elaborate with reference to the above example, two debugging technique that a programmer could use when handling errors. [4]
- f)** Currently working hours of contract staff are stored in a list of tuples with only the values of date and duration of each session. State two problems of the current implementation and suggest possible improvements. [3]

5. The following numbers are used to construct a binary search tree.

2 6 4 3 1 5 9

- a)** Construct a binary search tree using the above numbers inserting from left to right. [2]
- b)** Construct a binary search tree with the lowest height using the same set of numbers. [2]
- c)** Construct a binary search tree with the highest height using the same set of numbers. [1]
- d)** State the post order traversal of the BST of **a)**. [1]
- e)** State an advantage of using binary search tree over a hash table. [1]
- f)** Explain how you can reduce the height a binary search tree to minimum. State the time complexity of the transformation algorithm. [3]

6. Sorting and Searching

- a) Below shows a fixed set of integer hash keys.

[10, 31, 6, 76]

Explain clearly how you can use a hash table of size 7 to store all the keys above. You are to show the content of the hash table after all the keys are inserted. [4]

- b) Substitution cypher replaces letters with their numerical position in the English alphabet.

A=1 B=2 C=3 ... Z=26

So, for instance, using the cypher, the word KISS changes to 11-9-19-19. Give a reason why such substitution Cypher is a bad hash function. [1]

Study the bubble sorting algorithm below carefully and answer the questions from **6c) and 6d)**

```
def bubbleSort(lst):
    noSwaps = False
    while noSwaps == False:
        noSwaps = True
        for i in range(0, len(lst)-1):
            if lst[i] < lst[i+1]:
                noSwaps = False
                lst[i], lst[i+1] = lst[i+1], lst[i]
        print(lst)
```

- c) Trace the statement `print(lst)` when the following code is executed. [3]

```
>>> lst = [6,2,3,5,4]
>>> bubbleSort(lst)
```

- d) The bubble sort is modified. The recursive `onepass` function below performs a pass operation in a standard bubble sort. It returns `True` if swapping is performed during the pass and `False` otherwise. Complete the missing code **A**, **B** and **C**. [6]

```
def onepass(lst, i):
    if i > len(lst) - 2:
        return A
    else:
        if lst[i] > lst[i+1]:
            # No swap is needed to be performed.
            # So, recursively check on the next pair.
            return B
        else:
            # Perform swap on the i th and i+1 th item
            lst[i], lst[i+1] = lst[i+1], lst[i]
            return C

def bubbleSort(lst):
    swapped = True
    while swapped:
        swapped = False
        if onepass(lst, 0):
            swapped = True
```

A recursive procedure for finding a value, `SearchItem`, in an ordered array `X`, is as follows:

```
01 PROCEDURE Search(Low, High)
02   Middle <- (Low + High) // 2
03   IF X[Middle] = SearchItem
04     THEN OUTPUT "found"
05   ELSEIF X[Middle] > SearchItem
06     THEN
07       Search(Low, Middle - 1)
08     ELSE
09       Search(Middle + 1, High)
10   ENDIF
11 ENDPROCEDURE
```

- e) The Array `x` has 16 elements and the subscripts start at 1. State the calls to `Search` if the procedure is to search the array `x` for the value stored at `x[3]`. Your first call is `Search(1,16)`. [3]
- f) Explain why the search for an item does not present in the array would eventually result in the program crashing. [3]

7. The music school uses the following form for parents to book music lesson with the music teachers.

Lesson Booking Form					
Booking ID:	Humming_2022_001	Booking Date	20220507		
Start Date:	20220511	Number of Lessons:	10		
Customer ID:	Cust_02123	Customer Name:	Xiao Ming		
Customer Contact:	91234567	Customer Email:	xiaoming@gmail.com		
Teacher ID:	Teacher_007	Teacher Name:	James Bond		
Teacher Contact:	87614553	Teacher Email:	james@gmail.com		
Lesson No	Content	Instrument	Date	Time	Venue
1	Rhythm Intro 1	Drum	20220511	12:00	Classroom 01
2	Rhythm Intro 2	Drum	20220518	13:00	Classroom 01
3	Piano 1	Piano	20220525	12:30	Classroom 02
4	Piano 2	Piano	20220601	17:00	Classroom 01
...			...	...	...
Note for parent: <i>Lesson content and instrument for each lesson is fixed. However, for different student's booking, the date/time/venue of each lesson may vary.</i>					

- You are required to design a relational database solution for the company. Identify the tables that will give a normalized solution. Show the process of normalization. [6]
- Based on your solution in (a), draw a fully labelled ER diagram to show how the entities are related. [3]
- Explain the following terms and support with examples with reference to the above relational database model: [4]
 - Primary Key
 - Composite Key
- Explain with example, why relational database has advantage in maintaining data integrity comparing to a flat file solution. [3]
- A customer called in to the music school, requesting to change the date and time of the lesson. The manager managed to confirm a new slot, and hence need to contact both the customer and the teacher regarding this change through their contact number.

State and explain the tables and attributes needed to support the above scenario. [4]